Ronak Hingonia

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Summary — Software Engineer experienced in designing and building scalable web applications, specializing in cybersecurity.

Skills

Languages JavaScript, Python, Go
Frontend React.js, Astro, Tailwind, Redux
Backend Node.js, Express, RESTful APIs
Databases MongoDB, MySQL
Cybersecurity Cyrptography, Memory Safety, Web Security, Network Security
Cloud GCP
Others System Design, Computer Networks

Experience

CapIA.ai — **Mar 2025 – Apr 2025**
Back-End/Full Stack Developer

- Built a custom **Rule Engine** in **Python** integrated with **Vertex AI** and **GCP**, reducing **LLM API usage by 30–80%** and increasing **system efficiency by 65%**.
- Tested, benchmarked, and fine-tuned **10+ LLM models/prompts**, achieving a **35% improvement in response accuracy** and reducing **response time by 71%**.
- Implemented secure **OAuth login via Pennylane** in **Node.js** and extracted its **data-service logic** into a standalone **Python microservice** (managed via **Poetry**), enabling secure **user data synchronization**.
- Developed the **CapIA.ai website** from scratch with **Next.js**, implementing **SEO best practices**, **responsive design**, and improving **web performance by 30%** through modern **frontend optimizations**.

Apna College — **Aug 2024 – Nov 2024**
Teaching Assistant (MERN Stack)

- Resolved **10–14 daily queries** related to **coding issues**, **project troubleshooting**, and **explaining difficult topics**.
- Delivered **accurate and timely solutions**, improving student outcomes and earned a **4.7/5 TA rating**.

Education

Indian Institute of Technology Gandhinagar — **2020 – 2024**
B.Tech in Materials Science and Engineering
Minors: Computer Science and Engineering

Relevant Coursework
- Web Technologies
- Computer & Network Security
- Data Structures & Algorithms

Projects

botCam `How-To-Use-Guide` `Working-Flow` `Working-Demo.mp4` — Link
Convert phone camera into WebCam for PC

- Engineered a wireless phone-to-PC webcam system using **WebRTC** over **UDP**, establishing a secure **peer-to-peer (P2P) connection** for high-quality video streaming.
- Built a secure **WebSocket signaling server** in **Node.js** to enable device discovery and connection setup.
- **Tools & Technologies:** WebRTC, UDP, JavaScript, Node.js, Express, WebSockets, Google STUN, OpenSSL, Python

YouTube Clone `GitHub` `Architecture-Flow` — Link

- Built a full-stack video sharing platform with **Google Sign-In authentication**, automatic **video processing to 360p**, and seamless **video streaming**–delivering a YouTube-like experience **without using YouTube APIs**.
- Implemented a fully **serverless architecture** on **Google Cloud**, using automated workflows to process uploads, manage metadata, and deliver videos with zero manual intervention.
- **Tools & Technologies:** Next.js, Firebase, GCP (Cloud Functions, Cloud Storage, Cloud Run, Pub/Sub, Firestore), FFmpeg, Docker, TypeScript